

Figure 1: DHCP pool configuration on Router0 (show ip dhcp pool), showing the SALES, REPAIRS, ADMIN and PRINTER_ZONE pools, each with its own address range and 4 excluded addresses. The router is acting as the DHCP server with a separate scoped pool for each VLAN, matching the VLSM plan on 192.168.61.0/24. The Sales, Repairs and Admin pools each show 2 leased addresses, which are the two PCs in each department. PRINTER_ZONE shows 0 leases because the printer uses a static address (192.168.61.231).

```
Router0>enable
Router0#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.61.3     00D0.BA68.E877          --                      Automatic
192.168.61.2     0002.163A.674D          --                      Automatic
192.168.61.66    0001.4203.CD98          --                      Automatic
192.168.61.67    00D0.D3C7.2039          --                      Automatic
192.168.61.130   000C.CF29.1B12          --                      Automatic
192.168.61.131   00E0.F998.903A          --                      Automatic
Router0#
```

Figure 2:Each PC received an address from the pool for its own VLAN: 192.168.61.2 and .3 (Sales), .66 and .67 (Repairs), and .130 and .131 (Admin). Each lease is tied to the PC's MAC address, and the type is Automatic, which confirms the addresses came from DHCP

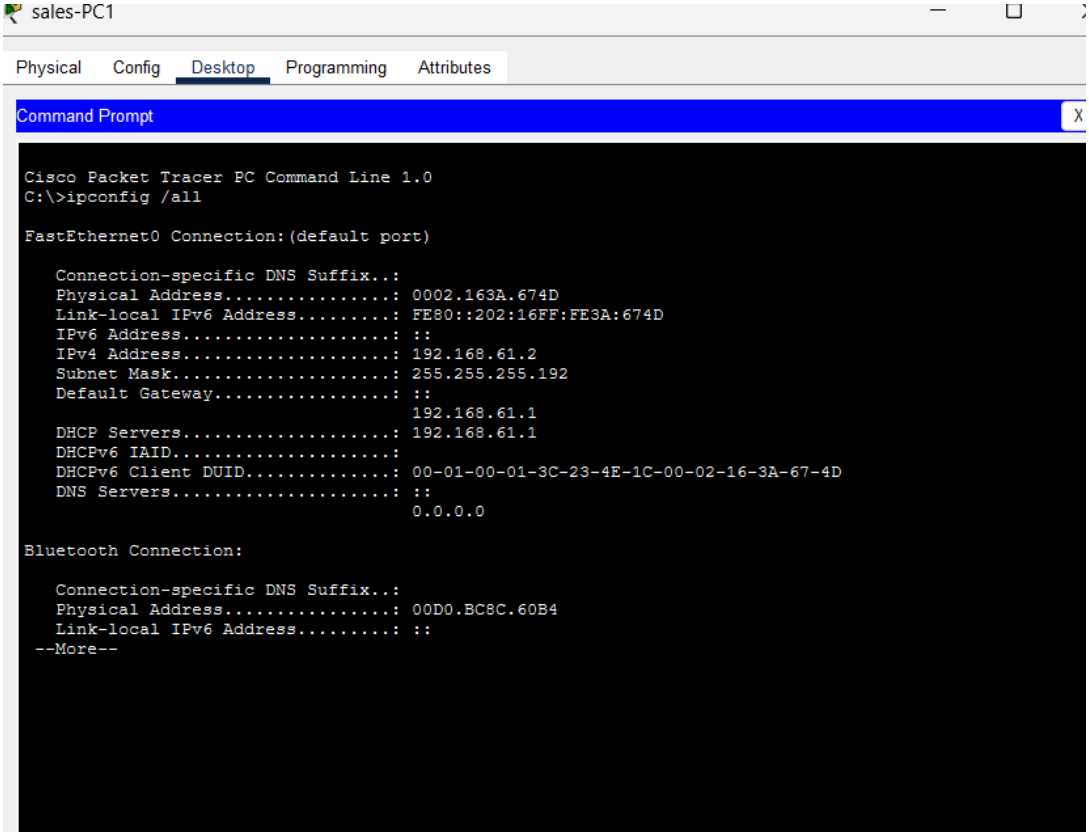


Figure 3: ipconfig /all on sales-PC1 (Sales VLAN), showing the address it received by DHCP.

sales-PC1 received 192.168.61.2 with subnet mask 255.255.255.192 (/26) and default gateway 192.168.61.1, which is the Router0 sub-interface for VLAN 10. The DHCP server is 192.168.61.1, so the router supplied the lease. The physical address 0002.163A.674D matches the binding for 192.168.61.2 in Figure 2.

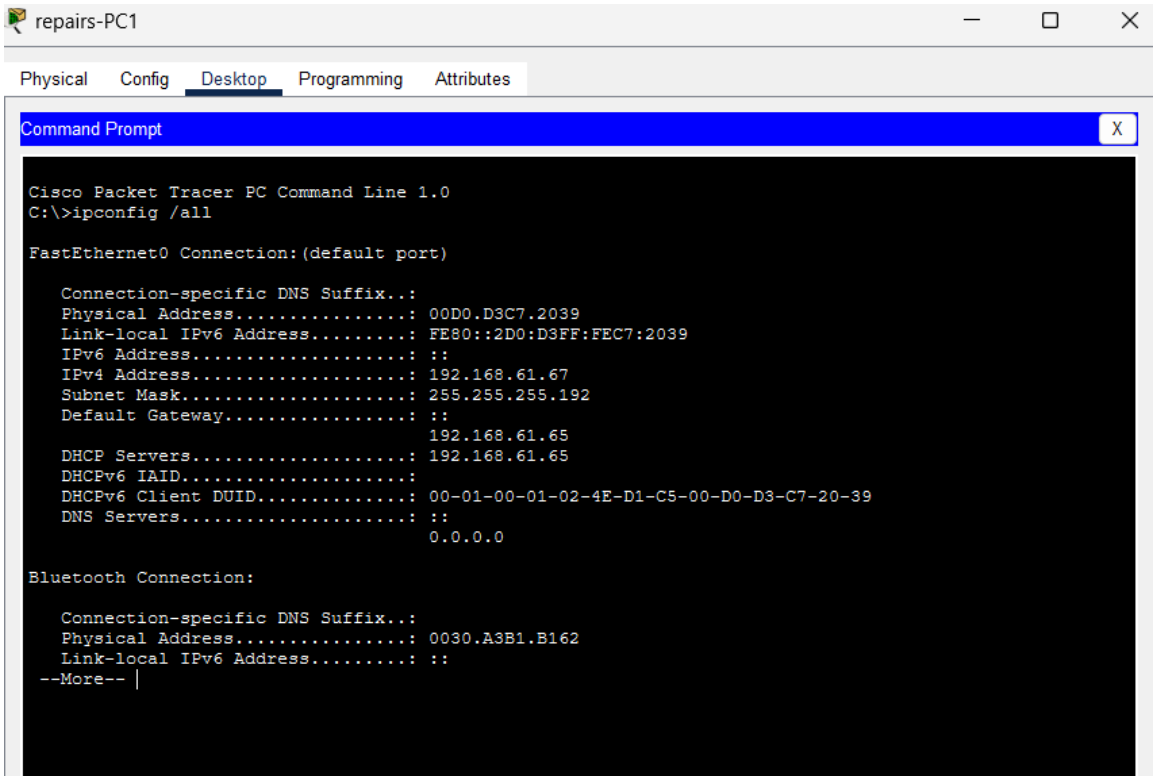


Figure 4: ipconfig /all on repairs-PC1 (Repairs VLAN), showing the address it received by DHCP.

repairs-PC1 received 192.168.61.67 with subnet mask 255.255.255.192 (/26) and default gateway 192.168.61.65, the Router0 sub-interface for VLAN 20. The DHCP server is 192.168.61.65, so the Repairs pool is served through its own VLAN's sub-interface. The physical address 00D0.D3C7.2039 matches the binding for 192.168.61.67 in Figure 2.

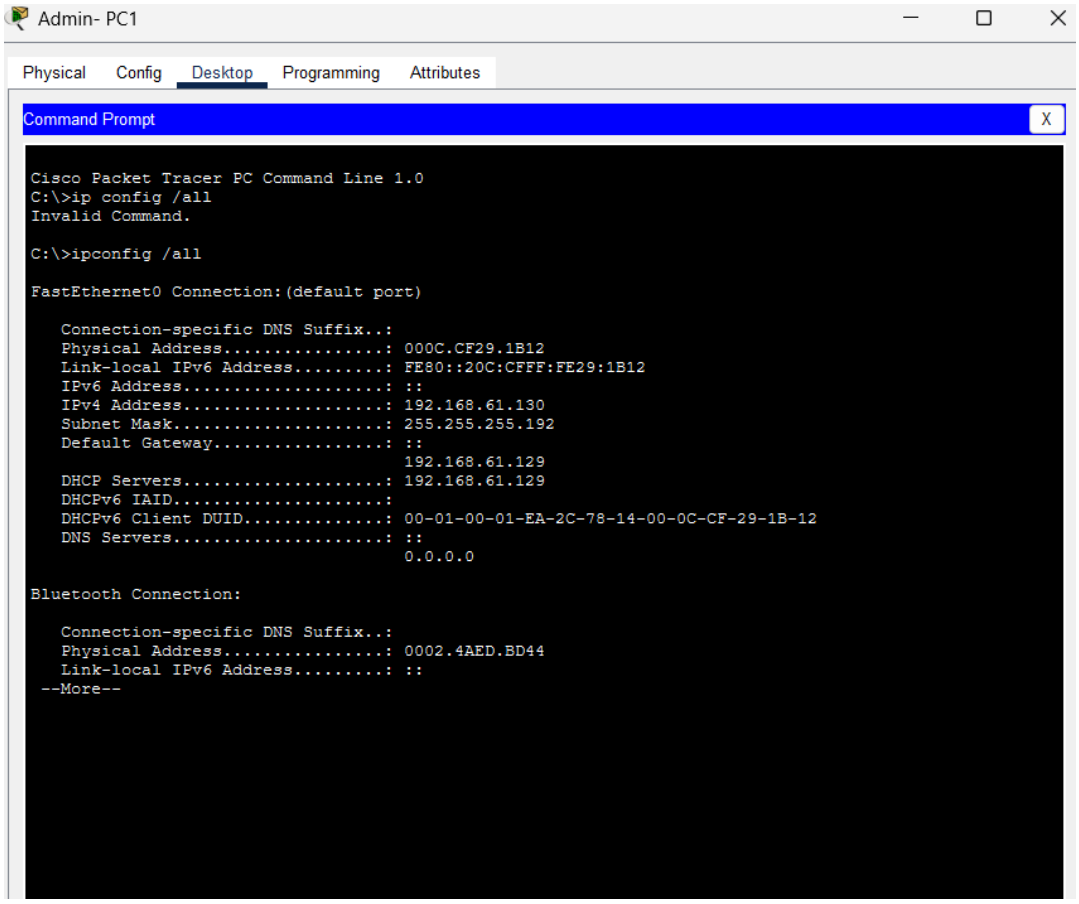


Figure 5: ipconfig /all on Admin-PC1 (Admin VLAN), showing the address it received by DHCP.

Admin-PC1 received 192.168.61.130 with subnet mask 255.255.255.192 (/26) and default gateway 192.168.61.129, the Router0 sub-interface for VLAN 30. The DHCP server is 192.168.61.129, so the Admin pool is served through its own sub-interface. The physical address 000C.CF29.1B12 matches the binding for 192.168.61.130 in Figure 2.

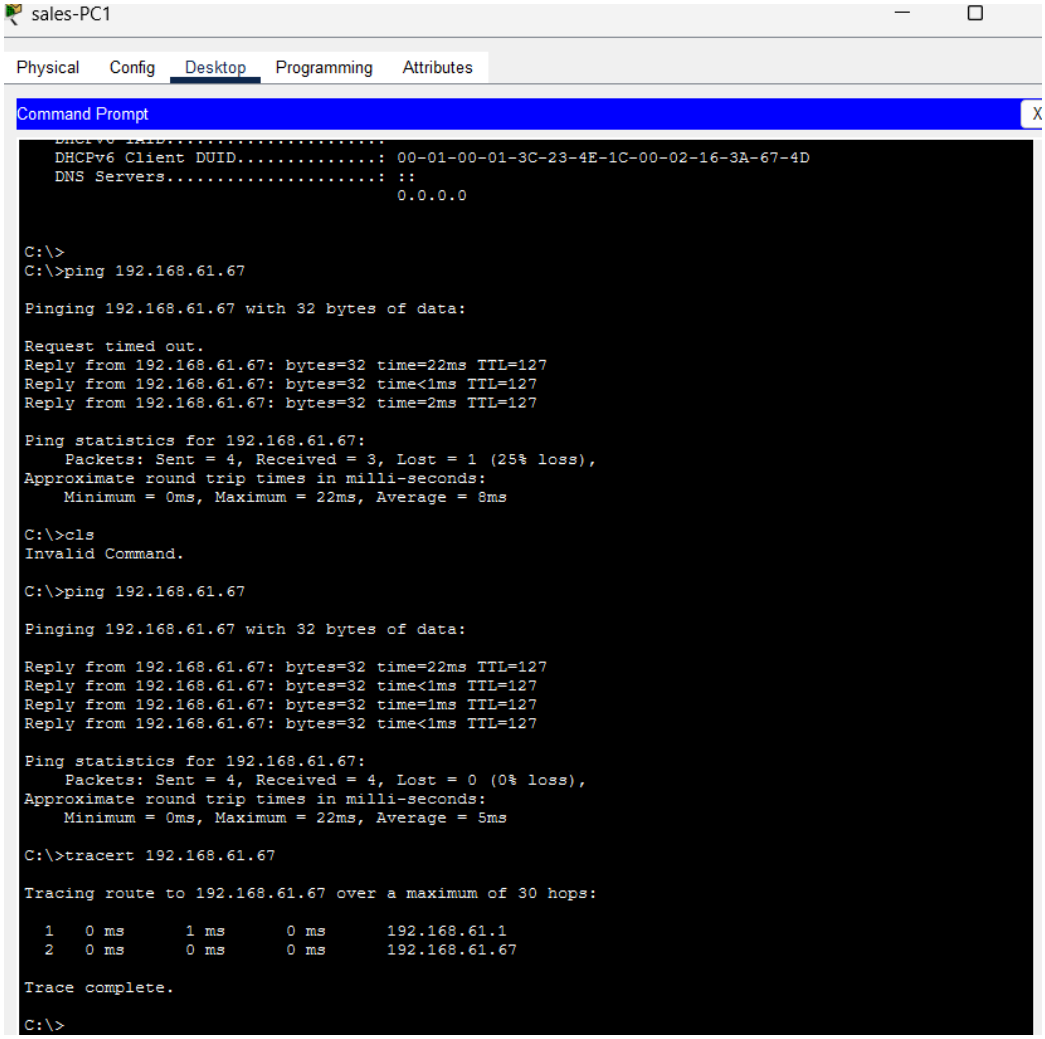


Figure 6: Ping and tracert from sales-PC1 (Sales VLAN) to repairs-PC1 (192.168.61.67, Repairs VLAN), testing inter-VLAN routing.

All 4 packets reached repairs-PC1 with 0% loss, and the TTL of 127 shows the packets crossed one router. The tracert confirms the path: hop 1 is 192.168.61.1, the Router0 Sales sub-interface, and hop 2 is the destination 192.168.61.67 in the Repairs VLAN. This shows Router-on-a-Stick is routing traffic between VLANs.

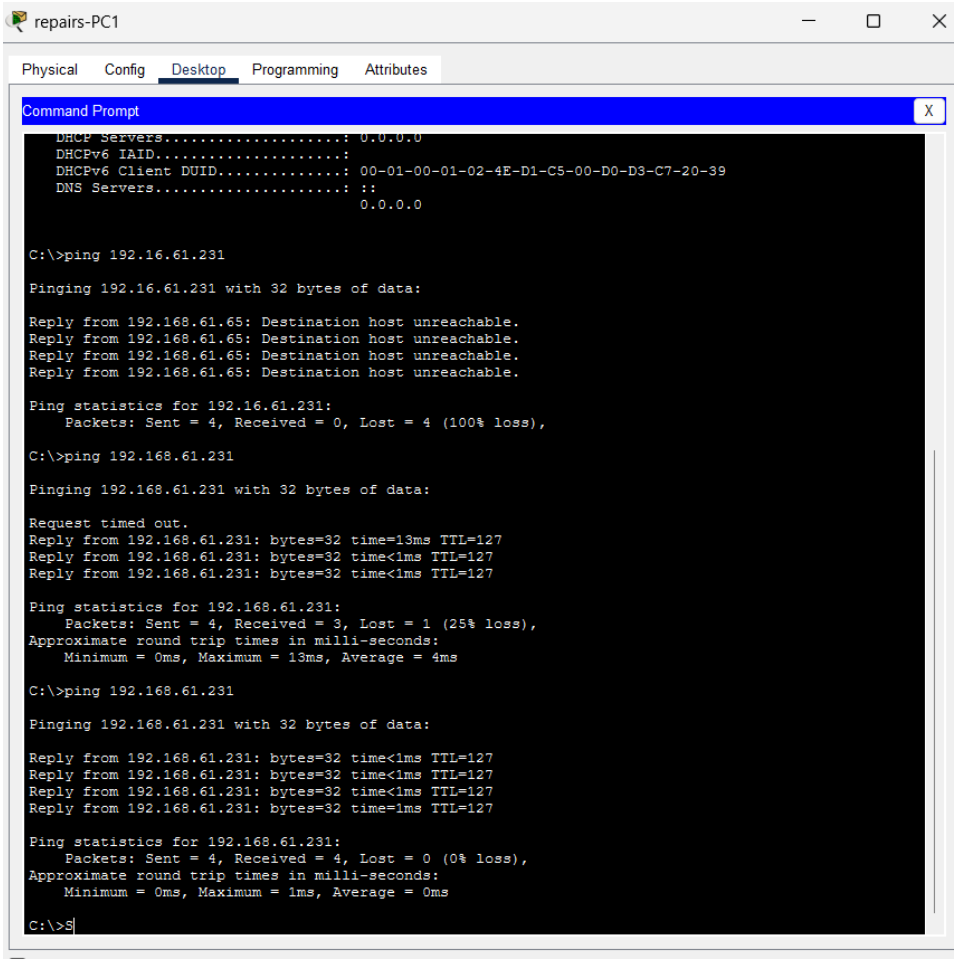


Figure 7: Ping from repairs-PC1 (Repairs VLAN) to the shared printer at 192.168.61.231 (Printer Zone, VLAN 50) .repairs-PC1 reached the shared printer in the Printer Zone. The TTL of 127 shows the traffic was routed through Router0, and the replies come from the printer's static address. This meets change request CR8, because the Repairs department can now print.

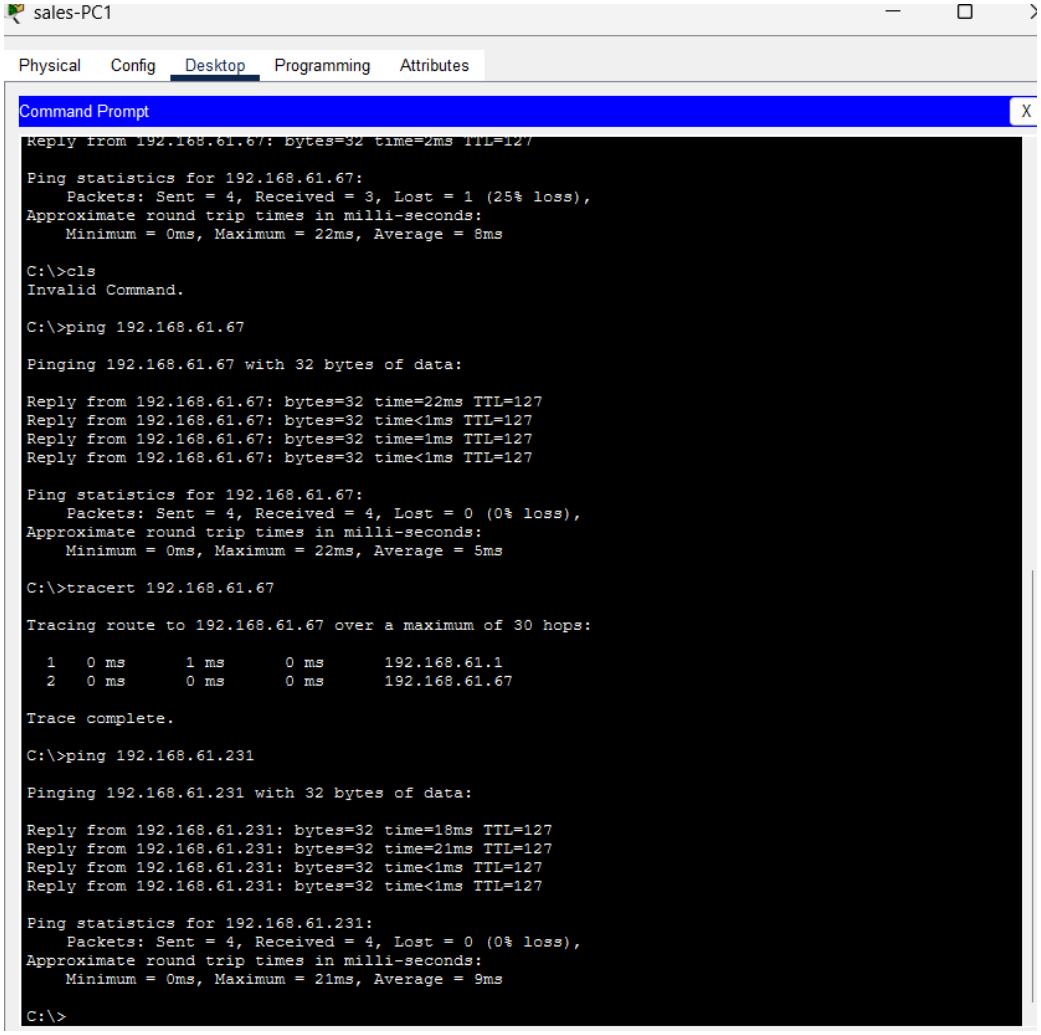


Figure 8: Ping from sales-PC1 (Sales VLAN) to the shared printer at 192.168.61.231 (Printer Zone, VLAN 50)

sales-PC1 reached the shared printer in the Printer Zone with 0% packet loss. The TTL of 127 shows the traffic was routed through Router0, and the replies come from the printer's static address. Together with Figure 7, this meets change request CR8: both the Sales and Repairs departments can now reach the shared printer.

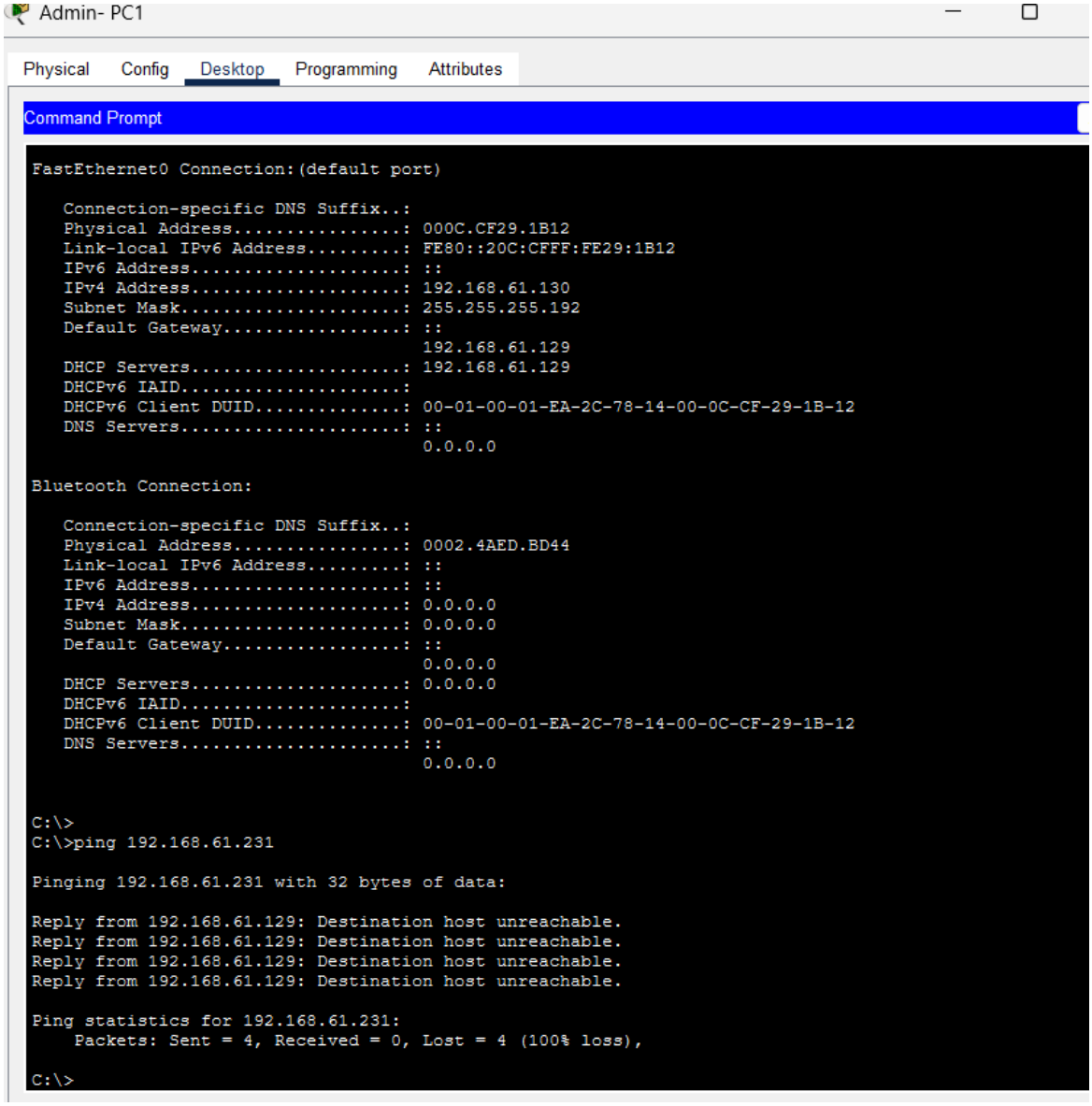


Figure 9: Ping from Admin-PC1 (Admin VLAN) to the shared printer at 192.168.61.231, blocked by the access list.

Admin-PC1 cannot reach the shared printer in the Printer Zone. All 4 packets were rejected with “Destination host unreachable”, returned by 192.168.61.129, the Admin gateway on Router0. This shows ACL ADMIN_NO_PRINTER_ZONE on Gig0/0.30 blocks the Admin VLAN from VLAN 50, as designed, while Sales and Repairs (Figures 7 and 8) can still print.

```
Router0>enable
Router0#show access-lists
Extended IP access list ADMIN_NO_PRINTER_ZONE
    10 deny ip 192.168.61.128 0.0.0.63 192.168.61.224 0.0.0.31 (4 match(es))
    20 permit ip any any (4 match(es))

Router0#SS
```

Figure 10: Access list ADMIN_NO_PRINTER_ZONE on Router0 (show access-lists), showing the deny rule that blocks the Admin VLAN from the Printer Zone

all IP traffic from the Admin subnet (192.168.61.128/26) to the Printer Zone subnet (192.168.61.224/27) is denied, and rule 20 permits all other traffic. The deny rule's 4 matches correspond to the 4 pings from Admin-PC1 in Figure 9, which confirms the ACL, not a fault, caused the failure. The permit rule keeps the Admin VLAN's access to every other network.